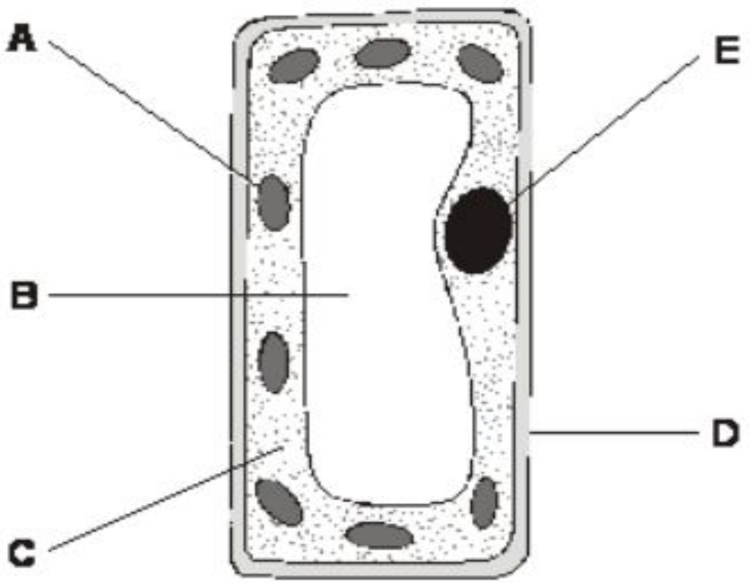
The diagram shows a plant cell.



Title – Food Tests Practical

Answer the exam questions in your books

(a) Give the name of part A.

........Chloroplasts..

Give the function of part A.

........photosynthesis..

....

2 marks

School

DO NOW

Food Tests Practical

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Complete Sam Learning Task

Due

We are learning how to:

- Test foods for starch, sugars, protein and fat.
- Predict the results of food tests for a range of foods

Keywords

starch

sugar

protein

fat

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CONNECTION

In the UK, the government is in charge of ensuring that all packaged foods are labelled to show what they contain and the different food groups in the food.

Accurate labelling allows people to make healthy choices about what they are eating and allows people with allergies to avoid certain foods.

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ACTIVATION

We can find out which food groups a food contains by testing it using chemicals. Scientists carry out the tests to allow foods to be labelled correctly. The tests include:

- **Starch**: a colour change with iodine solution from orange to blue/black.
- **Sugar**: a colour change with Benedict's solution from blue to orange.
- **Protein**: a colour change with Biuret solution from blue to mauve.
- **Fat**: appearance of a white milk-like solution when ethanol is added.

Food	Observations				Conclusion
	Test for starch	Test for sugar	Test for protein	Test for fats	
pasta	positive	negative	negative	negative	pasta contains starch
chicken	negative	negative	positive	positive	chicken contains protein
chickpeas	positive	negative	positive	negative	chickpeas contain protein and starch

Figure 1.2.3b: Testing food for starch

What do the results show about each of the foods

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ACTIVATION

Required practical 2: Food tests

RTL: investigate foods for the presence of glucose, protein, starch and lipids.

Apparatus

- Foods: potato, carrot, crisps, biscuits, cheese
- Spotting tile
- Pipette
- 10 cm^3 measuring cylinder or plastic syringe
- Beaker
- Boiling tubes
- Heating equipment
- Test tubes
- Iodine solution
- Benedict's reagent
- Biuret reagent
- Ethanol

Starter:

Copy RTL and apparatus

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ACTIVATION

Risk assessment

Hazard	Risk	Precaution

Risk assessment

Hazard	Risk	Precaution
Chemicals	Irritate skin/eyes/misuse	Gloves/goggles; correct labelling and handling
Hot water	Spillage/Burns/scalding	Keep hot water from edge of bench/correct handling/goggles

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ACTIVATION

Method: Food test

Food test	Method
Starch	1. Put a small piece of the food to test on a white tile. 2. Add two drops of iodine solution to the food. 3. If the iodine goes from yellow to blue-black, the food contains starch.
Glucose	1. Mix a small sample of the food with 3 cm3 of Benedict's solution in a boiling tube. 2. Heat the mixture in a hot water bath for three minutes. 3. If the solution goes a tomato-red colour, the food contains sugar.
Protein	1. Mix a small sample of the food with 3 cm3 of biuret solution. 2. Leave for two minutes. 3. If the mixture goes a pale purple colour, the food contains protein.
Lipid	1. Mix a small sample of the food with about 1 cm3 of ethanol in a dry test tube. 2. Pour the ethanol into a test tube full of cold water. 3. If the water goes milky white, the food contains lipids.

https://www.youtube.com/watch?v=akMLGbNA0gE

Method: Food test

Food test	**Method**
Starch	1. Put a small piece of the food to test on a white tile. 2. Add two drops of iodine solution to the food. 3. If the iodine goes from yellow to blue-black, the food contains starch.
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https://www.youtube.com/watch?v=akMLGbNA0gE

Results

Food being tested	Observations			
	Starch	Glucose	Protein	Lipid

Results

Food being tested	Observations			
	Starch	Glucose	Protein	Lipid

Results

Food being tested	Observations			
	Starch	Glucose	Protein	Lipid
Carrot	No	Yes	No	No
Cheese	No	No	Yes	Yes
Biscuit	Yes	No	No	Yes
Crisps	Yes	No	No	Yes
Potato	Yes	No	No	No

Conclusion

1.Which food groups were found in each of the foods you tested?

2. Were any of the colour changes difficult to see? Discuss your answer.

Some observations are difficult, as the food colour may affect the colour observed; biuret change can be difficult to spot.

3. Suggest any improvements you could make to the investigation.

Improvements could include crushing the food, dissolving the food in water first, holding the tubes against a white background to see the colour change more clearly

End of unit exam question guarantee

Required practical exam question

Georgji tested a sample for food for the presence of Protein and Starch. Explain how he did this? [4 marks]

Four relevant points needed

- **Test for protein uses biuret solution**
- **Positive result – purple/mauve**
- **Negative results - Blue**
- **Test for starch uses iodine**
- **Positive result – Blue/black**
- **Negative results – Brown/orange**

CONSOLIDATION

End of unit exam question guarantee

Mastery equations

1. **Write the equation for acid and metal oxides. [1 mark]**

   Acid + Metal Oxide □ Salt + Water

2. **Write the equation for acids and carbonates. [1 mark]**

   Acid + Metal Carbonate □ Salt + Water + Carbon Dioxide

3. **Write the equation for acids and metals. [1 mark]**

   Acid + Metal □ Salt + Hydrogen

3. **Write the equation for acids and bases. [1 mark]**

   Acid + Base □ Salt + Water

CONSOLIDATION

Match the food group to the chemical used to test for its presence.